ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)

ACADEX PRACTICE EXAMINATION

O-LEVEL · MATHEMATICS

MATHEMATICS 4004/2

PAPER 2 June 2023

2 hours 30 minutes

PAPER 2 — STRUCTURED QUESTIONS — CALCULATOR ALLOWED

Additional materials: Mathematical tables or a non-programmable calculator may be used.
INSTRUCTIONS: Section A: answer all questions (52 marks). Section B: answer any four of the seven questions (48 marks).

SECTION A (52 marks) — Answer ALL questions

1. 27 Form 4 pupils at St Mary's scored:<<
>>2 (8 pupils), 2 (4 pupils), 3 (5 pupils), 6 (4 pupils), 6 (6 pupils).<<
>>(a) Modal mark. [1]<<
>>(b) Mean mark. [4]<<
>>(c) Range. [2]<<
>>(d) One extra pupil scores 6. Effect on the mean? [2] **[9]**

2. AB ∥ CD with transversal EF in a street plan of Chitungwiza.<<
>>(a) Angle APQ = 42°. State the corresponding angle at Q, with a reason. [2]<<
>>(b) Find the co-interior angle to 42°. [2]<<
>>(c) A triangle has angles 42°, 46° and x°. Find x. [2]<<
>>(d) Classify the triangle (acute / right / obtuse). [2] **[8]**

3. A rectangular garden in Unit L measures 12 m by 10 m. A path of width 2 m runs all around the outside.<<
>>(a) Area of the garden. [2]<<
>>(b) Area of the path. [3]<<
>>(c) A circular bed of radius 7 m is later dug. Area of the bed, $\pi = 22/7$. [3] **[8]**

4. (a) Using ruler and compasses only, construct triangle ABC with AB = 7 cm, BC = 7 cm and AC = 8 cm. [4]<<
>>(b) Construct the perpendicular bisector of BC. [2]<<
>>(c) Mark the circumcentre and describe its property. [2] **[8]**

5. A cylindrical tank at St Mary's has radius 7 cm and height 20 cm. $\pi = 22/7$.<<
>>(a) Volume. [3]<<
>>(b) Curved surface area. [3]<<
>>(c) Total surface area including the base only (open top). [2] **[8]**

6. Line L: $y = 2x + -2$.<<
>>(a) Gradient and y-intercept. [2]<<
>>(b) Coordinates where L meets the x-axis. [2]<<
>>(c) Show A(0, -2) lies on L. [2]<<
>>(d) Equation of the line parallel to L through (0, 4). [2] **[8]**

SECTION B (48 marks) — Answer any FOUR of the seven questions

7. 50 pupils at St Mary's sat a test (0–50). An ogive is drawn.<<
>>(a) How to read the median from the ogive. [3]<<
>>(b) Median estimated as 34. Meaning? [2]<<
>>(c) How to find the IQR from the graph. [4]<<
>>(d) Why IQR is better than the range here. [3] **[12]**

8. OA = a = (6 ; 2), OB = b = (0 ; 2). M midpoint of AB.<<
>>(a) Vector AB. [3]<<
>>(b) |AB|. [3]<<
>>(c) Position vector OM. [3]<<
>>(d) Show AM = MB. [3] **[12]**

9. (a) y varies directly as x. When x = 4, y = 12. Find y when x = 10. [4]<<
>>(b) z varies inversely as x. When x = 3, z = 8. Find z when x = 6. [4]<<
>>(c) A farmer near Chitungwiza has at most 48 hours. Maize takes 2 h/ha, vegetables 1 h/ha. Inequality for hectares m and v. [4] **[12]**

10. (a) Solve $x^2 - 8x + 15 = 0$ by factorisation. [4]<<
>>(b) A rectangular plot in Chitungwiza is 5 m longer than it is wide. If the area is 24 m^2, form and solve an equation for the width w. [5]<<
>>(c) Sketch $y = (x - 3)(x - 5)$, showing intercepts. [3] **[12]**

11. Triangle P(2,1), Q(5,1), R(2,4).<<
>>(a) Reflect in the y-axis. Coordinates of P'. [4]<<
>>(b) Translate PQR by (3 ; -1). Give Q''. [3]<<
>>(c) Matrix (0 −1 ; 1 0) maps (x;y). Describe fully and find the image of Q. [5] **[12]**

12. A kombi leaves Chitungwiza at 08:00 and travels 120 km at constant speed, arriving after 2 hours. It stops 30 minutes, then returns at 80 km/h.<<
>>(a) Outward speed. [3]<<
>>(b) Time for the return. [3]<<
>>(c) Sketch a distance–time graph, labelling axes. [4]<<
>>(d) Average speed for the whole trip including the stop. [2] **[12]**

13. O is the centre. A, B, C on the circumference. AB is a diameter. Angle BAC = 44°.<<
>>(a) Angle ACB. Reason. [3]<<
>>(b) Angle BOC (centre, same arc BC). [3]<<
>>(c) D on the remaining circumference. Angle BDC. Reason. [3]<<
>>(d) State the angle-in-a-semicircle theorem. [3] **[12]**

END OF QUESTION PAPER

Worked solutions and mark-scheme notes are inside the ACADEX app (Extract & Study).